Hypothesis Testing with E-Values

Chapter 5: The Universal Inference E-Value

Based on Ramdas & Wang

Table of Contents

Notation you'll need I

We assume familiarity with e-values/e-variables and p-values from earlier chapters. Here is everything new in this chapter, defined plainly.

Composite hypothesis

A hypothesis is **simple** if it names a single distribution (e.g. $H_0 : p = N(0, 1)$). It is **composite** if it names a whole *set* of candidate distributions, $\mathcal{P}$ for the null or $\mathcal{Q}$ for the alternative (e.g. $H_0 : p = N(\mu, 1)$ for *some unknown* $\mu \in \mathbb{R}$). Composite hypotheses are the realistic case: we almost never know the true parameter value in advance.

Maximum likelihood estimator (MLE)

Given data $X_1, \ldots, X_m$ and a family of densities $\{p_\theta\}$, the MLE is the parameter value $\hat{\theta} = \arg\max_\theta \prod_{i=1}^{m} p_\theta(X_i)$ that makes the observed data as likely as possible. Equivalently it maximizes $\prod_i p(X_i)$ over p ranging over a set $\mathcal{P}$ of candidate densities (not necessarily indexed by a Euclidean parameter).

Notation you'll need II

"Irregular model" (plain meaning)

Classical hypothesis testing theory (Wilks' theorem, asymptotic normality of the MLE, chi-squared limits for likelihood ratios) relies on *regularity conditions*: e.g. the number of parameters is fixed and small, the true parameter is an interior point, and the Fisher information matrix is invertible. A model is **irregular** when one or more of these fail – for instance:

- a parameter is only identifiable under the alternative but not under the null (or vice versa),
- the true parameter sits on the boundary of the parameter space,
- the number of parameters grows with the sample size,
- the log-likelihood ratio statistic is not asymptotically χ^2.

In irregular models, standard asymptotic tests may simply not have any known, provably valid version – even approximately, even as $n \to \infty$.

Notation you'll need III

Exchangeable data

A sequence $Z_1, Z_2, \ldots$ is **exchangeable** if the joint distribution of any finite subvector $(Z_1, \ldots, Z_n)$ is unchanged under any reordering (permutation) of the indices. iid data is exchangeable; so are repeated copies of the same random variable. Exchangeability is weaker than iid-ness but strong enough to support powerful concentration-free inequalities, as we'll see in Section 5.4.

The universal inference idea, in one sentence

If you can compute (or upper bound) the maximum likelihood under the null hypothesis, you can turn that single computation into a valid e-variable – and hence a valid test – for a composite null against a composite alternative, with *no regularity conditions whatsoever*.

5.1 Motivation: irregular models I

Why do we need a new tool at all? For simple nulls we already know how to build e-variables (earlier chapters): just use a likelihood ratio against a fixed alternative. The trouble starts when *both* the null and the alternative are composite – then there is no unique “the” likelihood ratio, and classical theory (based on asymptotics of the MLE) may not even apply.

Motivating example 1: testing for a mixture

Let $p_{\mu_1,\mu_2,\lambda}$ be the density of $(1-\lambda)N(\mu_1,1)+\lambda N(\mu_2,1)$ (a two-component Gaussian mixture). Consider testing

$$H_0 : \lambda = 0 \quad \text{(equivalently } \mu_1 = \mu_2\text{)} \qquad \text{vs.} \qquad H_1 : \text{unrestricted (except } \neq H_0\text{)}.$$

This is a genuinely composite null (any $\mu_1 = \mu_2$ value is allowed) versus a genuinely composite alternative. It is also *irregular*: under H_0 the parameter μ_2 is not identified (any value of μ_2 gives the same null distribution once $\lambda = 0$), and under H_1, λ is not identified as $\mu_1 \to \mu_2$. This breaks the standard machinery (no χ^2 limit for the likelihood ratio test), and to the authors’ knowledge, universal inference gives the first test here with a provable type-I error guarantee.

5.1 Motivation: irregular models II

Motivating example 2: shape constraints

A density p on $\mathbb{R}$ is **log-concave** if $p = e^g$ for some concave function g (log-concavity is a common, flexible nonparametric shape assumption – it includes the Gaussian, the Laplace, the uniform, etc., but excludes heavy-tailed or strongly multimodal densities). Consider

$$H_0 : p \text{ is log-concave} \qquad \text{vs.} \qquad H_1 : p \text{ is not log-concave.}$$

Both hypotheses are composite (they are sets of densities, not single densities), Θ is infinite-dimensional, and no other provably valid test for this problem is known. Universal inference applies directly, and is computationally efficient here because the log-concave MLE can be computed in polynomial time.

The universal-inference reduction

Universal inference is a *reduction*: it turns the (hard) problem of hypothesis testing under a composite null and alternative, with no regularity conditions, into the (often easier) problem of computing – or just upper bounding – the **maximum likelihood under the null**.

5.1 Motivation: irregular models III

Two standing assumptions for this chapter:

1. $\mathcal{P}$ (null) and $\mathcal{Q}$ (alternative) share a common density scale, so every distribution can be represented by an ordinary density $p(x)$ or $q(x)$.
2. The data is $X^n = (X_1, \ldots, X_n)$ with iid entries; we identify a distribution $\mathbb{P} \in \mathcal{P}$ with the density $p(x)$ of a single coordinate, and use $p \in \mathcal{P}$ and $\mathbb{P} \in \mathcal{P}$ interchangeably.

5.2 Intuition: why not just use the likelihood ratio directly?

The obstacle. For a simple null p vs. a simple alternative q, the likelihood ratio $q(X)/p(X)$ is already a valid e-variable: $\mathbb{E}^p[q(X)/p(X)] = \int q(x)\,dx = 1$. So why not just plug in the MLE $\hat{p}$ under the (composite) null and the MLE $\hat{q}$ under the alternative, and use $\hat{q}(X^n)/\hat{p}(X^n)$?

The problem

Under a composite null, we don't know the true $p \in \mathcal{P}$. If we estimate $\hat{p}$ *using the very same data* we test on, the ratio $\hat{q}(X^n)/\hat{p}(X^n)$ is no longer guaranteed to have expectation ≤ 1 – $\hat{p}$ was chosen (in part) to explain that data, so it artificially inflates the likelihood ratio in its own favor. This double-use of the data breaks the e-variable property.

The fix: split the data

Split the sample into two independent halves. Use one half only to *choose* a plausible alternative distribution. Use the other half only to *fit the null MLE and evaluate the ratio*. Because the null MLE is honestly maximizing likelihood over the data it is tested against – and only over that data – the key inequality $\prod \hat{p}_0(X_i) \geq \prod p(X_i)$ becomes available, and it is exactly what restores the ≤ 1 expectation guarantee. This is the single trick behind the whole chapter.

Step-by-step: building the split likelihood ratio e-variable

Step 1 – Split the data

Divide $X^n = (X_1, \ldots, X_n)$ into two *independent* sets D_0 and D_1. E.g. let $I \subset [n]$ be a random subset independent of X^n (flip an independent coin for each point), and set $D_0 = \{X_i : i \in I\}$, $D_1 = \{X_i : i \notin I\}$. Roughly equal sizes are a reasonable default, but validity does *not* depend on the split proportion (only e-power does). The split need not even be random: $D_0 = \{X_1, \ldots, X_m\}$, $D_1 = \{X_{m+1}, \ldots, X_n\}$ works too.

Step 2 – Pick a candidate alternative using D_1

Choose any $\hat{q}_1 \in \mathcal{Q}$ based only on D_1: an MLE, a Bayes estimator (posterior mean/mode under some prior), or a robust estimator. This choice is *unrestricted* – it cannot affect validity, only power/e-power.

Step 3 – Fit the null MLE using D_0

$$\hat{p}_0 = \arg\max_{p \in \mathcal{P}} \prod_{i \in D_0} p(X_i).$$

Here $\hat{p}_0$ is the density in the null family $\mathcal{P}$ that best explains D_0 alone.

The split likelihood ratio e-variable

Step 4 – Form the ratio, evaluated on D_0

$$E \;=\; \prod_{i \in D_0} \frac{\hat{q}_1(X_i)}{\hat{p}_0(X_i)}. \qquad (5.1)$$

Symbols: D_0, D_1 are the two (independent) halves of the data; $\hat{q}_1$ is the alternative fitted *only* using D_1; $\hat{p}_0$ is the null MLE fitted *only* using D_0; the product (equivalently, the ratio of the two "likelihoods") is evaluated over the points in D_0.

We call E the **split likelihood ratio e-variable**. Note the asymmetric roles: D_1 only supplies the fixed candidate $\hat{q}_1$; D_0 is used twice – once to fit $\hat{p}_0$, and once again (the same data!) to evaluate the ratio.

Variant: cross-fit likelihood ratio e-variable

Swap the roles of D_0, D_1, recompute E^{swap} by the same recipe, and average:

$$E^{\text{cross-fit}} = \frac{E + E^{\text{swap}}}{2}.$$

Combined with Markov's inequality (Chapter 2) this yields the *split LRT* or *cross-fit LRT*: a level-α test that rejects H_0 when the e-variable exceeds $1/\alpha$.

Theorem 5.1: validity of the split LR e-variable I

Theorem 5.1

The split likelihood ratio statistic E defined in (5.1) is an e-variable for $\mathcal{P}$: $\mathbb{E}^{\mathbb{P}}[E] \leq 1$ for every $\mathbb{P} \in \mathcal{P}$.

Why this deserves its own theorem: it is what lets us plug E into Markov's inequality and get a valid level-α test, no matter how irregular the model is, and no matter what estimator was used for $\hat{q}_1$.

Theorem 5.1: validity of the split LR e-variable II

Proof, step by step

1. By definition, $\hat{p}_0$ maximizes $\prod_{i \in D_0} p(X_i)$ over $p \in \mathcal{P}$, so for *any* $\mathbb{P} \in \mathcal{P}$,

$$\prod_{i \in D_0} \hat{p}_0(X_i) \;\geq\; \prod_{i \in D_0} p(X_i).$$

2. Dividing through (the numerator $\hat{q}_1(X_i)$ is common) and taking $\mathbb{E}^{\mathbb{P}}[\cdot \mid D_1]$:

$$\mathbb{E}^{\mathbb{P}}\left[\prod_{i \in D_0} \frac{\hat{q}_1(X_i)}{\hat{p}_0(X_i)} \,\middle|\, D_1\right] \;\leq\; \mathbb{E}^{\mathbb{P}}\left[\prod_{i \in D_0} \frac{\hat{q}_1(X_i)}{p(X_i)} \,\middle|\, D_1\right] = 1.$$

3. The final equality holds because, conditional on D_1, $\hat{q}_1$ is a *fixed* density, $D_0 \perp D_1$, and a likelihood ratio of any fixed alternative to the true density p is an exact e-variable with mean exactly 1.
4. Take a further expectation over D_1 and use the tower property: $\mathbb{E}^{\mathbb{P}}[E] = \mathbb{E}^{\mathbb{P}}\left[\mathbb{E}^{\mathbb{P}}[E \mid D_1]\right] \leq 1$. ■

Remark: mixture universal inference

Remark 5.2 (mixture variant)

Instead of plugging in a single point estimate $\hat{q}_1$, use D_1 to choose a whole distribution ν over $\mathcal{Q}$ (e.g. a Bayesian posterior), and define

$$E = \int_{\mathcal{Q}} \prod_{i \in D_0} \frac{q(X_i)}{\hat{p}_0(X_i)} \, \nu(dq).$$

Intuition: the null side still uses the plug-in MLE $\hat{p}_0$, but the alternative side is now an *average* (mixture) over candidate alternatives rather than a single plug-in estimate. Since an average of e-variables is again an e-variable (each $\prod_{i \in D_0} q(X_i)/\hat{p}_0(X_i)$ is itself a valid e-variable conditional on D_1, by the same argument as Theorem 5.1), the mixture version is valid by an almost identical proof – integrals are just (possibly uncountable) averages. In practice, a prior with mass spread over all of $\mathcal{Q}$ tends to help e-power, especially in small samples, and can be viewed simply as a smoothed version of the plug-in MLE.

Running example: Gaussian vs. Gaussian mixture

Setup. H_0: $N(\mu, 1)$ (one unknown mean). H_1: 2-component mixture $(1-\lambda)N(\mu_1, 1) + \lambda N(\mu_2, 1)$, μ_1, μ_2, λ all free (the irregular model of Sec. 5.1). We simulate $n = 40$ points from a genuine mixture, split $20/20$ into D_0, D_1.

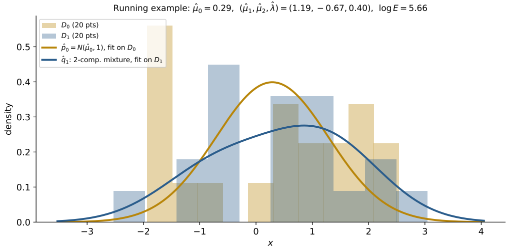
What we computed (one concrete run)

- Null MLE on D_0 (sample mean, var. known $= 1$): $\hat{\mu}_0 = 0.289$.
- Alt. fit on D_1 (2-comp. mixture MLE, numerical opt.): $\hat{\mu}_1 = 1.194$, $\hat{\mu}_2 = -0.667$, $\hat{\lambda} = 0.403$.
- Split LR e-variable, evaluated on D_0:

$$\log E = \sum_{i \in D_0} \Big[\log \hat{q}_1(X_i) - \log \hat{p}_0(X_i) \Big] = 5.655,$$

i.e. $E \approx 285.8$.

Since $E = 285.8 \gg 1/\alpha = 20$ at $\alpha = 0.05$, the split LRT rejects H_0 here – correctly, since the data are a mixture.



Python: split LR e-variable (5.2) – part I

This reproduces the numbers on the previous slide: null MLE (sample mean) on D_0, a 2-component Gaussian mixture MLE on D_1 (via numerical optimization), and the split LR e-variable.

```python
import numpy as np
from scipy.stats import norm
from scipy.optimize import minimize

def mixture_negloglik(params, x):
    mu1, mu2, lam = params
    lam = min(max(lam, 1e-6), 1 - 1e-6)
    dens = (1-lam)*norm.pdf(x, mu1, 1) + lam*norm.pdf(x, mu2, 1)
    return -np.sum(np.log(np.clip(dens, 1e-300, None)))

def split_LR_evariable(D0, D1):
    # Step 2: alternative fit on D1 (mixture MLE)
    res = minimize(mixture_negloglik, x0=[-1, 1, 0.5], args=(D1,),
                   method="Nelder-Mead")
    mu1_hat, mu2_hat, lam_hat = res.x
    # Step 3: null MLE on D0 (single Gaussian mean, var=1)
    mu0_hat = D0.mean()
    # Step 4: ratio, evaluated on D0
    q1 = (1-lam_hat)*norm.pdf(D0, mu1_hat, 1) + lam_hat*norm.pdf(D0, mu2_hat, 1)
    p0 = norm.pdf(D0, mu0_hat, 1)
    logE = np.sum(np.log(q1) - np.log(p0))
    return logE, np.exp(logE)
```

Python: split LR e-variable (5.2) – part II

Generating the $n = 40$ toy dataset, splitting it $20/20$, and calling the function defined on the previous slide:

```
rng = np.random.default_rng(2024)
comp = rng.random(40) < 0.5
x = np.where(comp, rng.normal(1.2, 1, 40), rng.normal(-1.2, 1, 40))
idx = rng.permutation(40)
D0, D1 = x[idx[:20]], x[idx[20:]]
logE, E = split_LR_evariable(D0, D1)
print(f"log E = {logE:.3f}, E = {E:.3e}")    # matches the slide above
```

5.3 Intuition: taming the randomness of splitting

The obstacle. The split LR e-variable introduces an extra source of randomness that has nothing to do with the data: the *coin flips* used to decide the split. Two analysts with the exact same data X^n but different random splits can get quite different values of E, and hence reach different conclusions. This "algorithmic randomness" is unsatisfying and can hurt e-power.

The fix: split many times and average

Recompute the split LR e-variable independently B times, using B fresh, iid random splits of the same data (these can all be computed in parallel), giving e-variables $E^{(1)}, \ldots, E^{(B)}$. Average them:

$$\bar{E}^{(B)} = \frac{E^{(1)} + \cdots + E^{(B)}}{B}. \qquad (5.2)$$

We call $\bar{E}^{(B)}$ the **subsampled likelihood ratio e-variable**. This is a form of *Rao–Blackwellization*: we remove variance introduced by an artificial randomization device, by averaging over that device.

As a special case, $B = 2$ with the two splits swapped recovers the *cross-fit* likelihood ratio e-variable from Section 5.2.

Proposition 5.3: averaging can only help

Proposition 5.3 (improving e-power by averaging)

Let $E^{(1)}, \dots, E^{(B)}$ be *arbitrary* e-variables (not necessarily split LR ones), and $\bar{E}^{(B)}$ their average. For any distribution $\mathbb{Q}$,

$$\mathbb{E}^{\mathbb{Q}}\big[\log \bar{E}^{(B)}\big] \;\geq\; \frac{1}{B}\sum_{b=1}^{B} \mathbb{E}^{\mathbb{Q}}\big[\log E^{(b)}\big],$$

provided the right side is well defined. In particular, if $E^{(1)}, \dots, E^{(B)}$ are iid copies of the split LR e-variable (5.1), the subsampled e-variable (5.2) has e-power (against any $\mathbb{Q}$) at least as large as that of a single split LR e-variable.

Proof idea: $\log(\cdot)$ is a concave function, so by Jensen's inequality $\log\big(\frac{1}{B}\sum_b E^{(b)}\big) \geq \frac{1}{B}\sum_b \log E^{(b)}$ pointwise; take $\mathbb{E}^{\mathbb{Q}}$ of both sides. That's it – the whole argument is one line of concavity. As $B \to \infty$, $\bar{E}^{(B)}$ converges almost surely to a fixed limiting random variable (this uses the backward-martingale fact of Section 5.4).

The subsampled LRT: a sequential procedure

Subsampled LRT

For $b \geq 1$, let $\bar{E}^{(b)} = (E^{(1)} + \cdots + E^{(b)})/b$. The subsampled LRT **rejects** H_0 as soon as any running average $\bar{E}^{(b)}$ exceeds $1/\alpha$, for any $b \geq 1$. One can compute $E^{(1)}, E^{(2)}, \ldots$ one split at a time, updating the running average $\bar{E}^{(1)}, \bar{E}^{(2)}, \ldots$ sequentially, and stop (rejecting) at the first time any of these exceeds $1/\alpha$. Crucially, B can be *data-dependent* (e.g. "keep splitting until convinced, or until compute budget runs out") *without* violating the type-I error guarantee.

Proposition 5.4

The subsampled LRT controls type-I error at level α, and is at least as powerful as the (single-split) split LRT.

The power claim is immediate: the very first step of the subsampled procedure, $b = 1$, is exactly the split LRT. The type-I error claim is *not* obvious – naively, "keep testing until you reject" looks like it should inflate type-I error (optional stopping). It doesn't here, and the reason is the **exchangeable Markov inequality** of Section 5.4, coming up next.

Python: subsampled / cross-fit LR e-variable (5.3) – part I

A running average of independently split LR e-variables, stopped the first time it crosses $1/\alpha$ (sequential subsampled LRT).

```python
import numpy as np

def subsampled_LRT(x, alpha, B_max, split_lr_fn, rng):
    """x: full data array. split_lr_fn(D0, D1, rng) -> log E for one split.
    Returns (rejected: bool, b_stop: int or None, running_Ebar: list)."""
    n = len(x)
    logEs, Ebars = [], []
    for b in range(1, B_max + 1):
        idx = rng.permutation(n)
        half = n // 2
        D0, D1 = x[idx[:half]], x[idx[half:]]
        logEs.append(split_lr_fn(D0, D1, rng))
        Ebar = np.mean(np.exp(logEs))          # average of E's, not of logE's
        Ebars.append(Ebar)
        if Ebar >= 1 / alpha:                  # exchangeable Markov guarantee
            return True, b, Ebars              # kicks in here -- see Sec. 5.4
    return False, None, Ebars
```

Python: subsampled / cross-fit LR e-variable (5.3) – part II

Wrapping the split-LR function from Section 5.2 and running the sequential subsampled LRT on the toy dataset:

```
# split_lr_fn wraps split_LR_evariable() from Section 5.2 and
# returns only its logE.
rng = np.random.default_rng(1)
split_lr_fn = lambda D0, D1, rng: split_LR_evariable(D0, D1)[0]
rejected, b_stop, Ebars = subsampled_LRT(x, alpha=0.05, B_max=20,
                                          split_lr_fn=split_lr_fn, rng=rng)
print("rejected:", rejected, " at split #", b_stop)
```

5.4 Intuition: why do we need a new inequality?

The obstacle. Ordinary Markov's inequality, $\mathbb{P}(Z \geq 1/\alpha) \leq \alpha\,\mathbb{E}[Z]$ for $Z \geq 0$, is a *one-shot* guarantee: it bounds the chance of a single random variable being large. But the subsampled LRT looks at a whole *running average* $\bar{E}^{(1)}, \bar{E}^{(2)}, \ldots$ and rejects if *any* of them ever crosses the threshold – this is an "optional stopping"-flavored claim, and plain Markov does not cover it.

The fix: exploit exchangeability

$E^{(1)}, E^{(2)}, \ldots$ (the repeated, independent-split e-variables) are not just any random variables: they are **exchangeable** – reordering the splits doesn't change their joint distribution, since the splits are iid. It turns out exchangeability alone (no independence needed) is exactly enough structure to control the *whole trajectory* of running averages, via a *backward martingale* argument.

Definition: exchangeable sequence

$Z_1, \ldots, Z_n$ are exchangeable if $(Z_1, \ldots, Z_n) \overset{d}{=} (Z_{\pi(1)}, \ldots, Z_{\pi(n)})$ for every permutation π of $\{1, \ldots, n\}$. An infinite sequence is exchangeable if every finite subsequence is.

Fact 5.5 and Theorem 5.6 (EMI) I

Fact 5.5 (backward martingale)

For any exchangeable, integrable sequence $Z_1, Z_2, \ldots$, the running-mean process $\bar{Z}_n := (Z_1 + \cdots + Z_n)/n$ is a **backward martingale**: with $\mathcal{G}_{n+1} := \sigma(\bar{Z}_{n+1}, \bar{Z}_{n+2}, \ldots)$ (information from index $n+1$ onward),

$$\mathbb{E}^{\mathbb{P}}[\bar{Z}_n \mid \mathcal{G}_{n+1}] = \bar{Z}_{n+1} \quad \text{for all } n.$$

Moreover $\bar{Z}_n$ converges almost surely to an integrable random variable as $n \to \infty$ (this is Doob's backward martingale convergence theorem).

Fact 5.5 and Theorem 5.6 (EMI) II

Theorem 5.6: Exchangeable Markov Inequality (EMI)

For any exchangeable sequence $Z_1, Z_2, \ldots$ and any $\alpha > 0$,

$$\mathbb{P}\left(\sup_{n\geq 1}\left|\frac{1}{n}\sum_{i=1}^{n} Z_i\right| \geq \frac{1}{\alpha}\right) \leq \alpha\,\mathbb{E}^{\mathbb{P}}[|Z_1|]. \qquad (5.3)$$

In particular, if $Z_1, Z_2, \ldots$ are e-variables,

$$\mathbb{P}\left(\exists n \geq 1 : \frac{1}{n}\sum_{i=1}^{n} Z_i \geq \frac{1}{\alpha}\right) \leq \alpha.$$

Symbols: $\sup_{n\geq 1}$ – the *worst case over all sample sizes ever looked at*, not just one fixed n; this is exactly the "look at the whole trajectory" guarantee the subsampled LRT needs. The proof combines Fact 5.5 with a time-reversed Ville's inequality (Chapter 7).

Turning arbitrary dependence into exchangeability I

Why this matters: EMI needs exchangeability, but many real datasets are only iid, or not even that. A simple trick – random permutation – manufactures exchangeability out of thin air.

Corollary: random-permutation trick

Let $Z_1, \ldots, Z_n$ be *arbitrarily dependent* (not necessarily exchangeable), let π be an independent uniformly random permutation of $\{1, \ldots, n\}$. Then for any $\alpha > 0$,

$$\mathbb{P}\left(\sup_{1 \le m \le n} \frac{1}{m} \sum_{i=1}^{m} |Z_{\pi(i)}| \ge \frac{1}{\alpha} \right) \le \alpha \, \frac{\mathbb{E}^{\mathbb{P}}[|Z_1| + \cdots + |Z_n|]}{n}.$$

Randomly permuting the (otherwise arbitrary) sequence makes it exchangeable *by construction*, so (5.3) applies to $Z_{\pi(1)}, \ldots, Z_{\pi(n)}$.

Turning arbitrary dependence into exchangeability II

Variant: sampling from a bag

Take N arbitrarily dependent random variables, put them in a bag, and draw $n \leq N$ of them uniformly at random (with or without replacement) as $Z_{\pi(1)}, \ldots, Z_{\pi(n)}$. Then

$$\mathbb{P}\left(\sup_{1\leq m\leq n} \frac{1}{m}\sum_{i=1}^{m} |Z_{\pi(i)}| \geq \frac{1}{\alpha}\right) \leq \alpha\, \frac{\mathbb{E}^{\mathbb{P}}[|Z_1| + \cdots + |Z_N|]}{N}.$$

The random sampling process itself induces the exchangeability needed to invoke (5.3), even though the original bag of N variables need not be exchangeable.

Theorem 5.7: Exchangeable Uniformly Randomized MI (EUMI)

Theorem 5.7 (EUMI)

Let $Z_1, \ldots, Z_n$ be exchangeable. Then for any $a \in (0, 1)$,

$$\mathbb{P}\left(Z_1 \geq U/a \text{ or } \exists t \leq n : \left| \sum_{i=1}^{t} Z_i / t \right| \geq 1/a \right) \leq a \cdot \mathbb{E}^{\mathbb{P}}[|Z_1|],$$

where U is a uniform random variable on $[0, 1]$, independent of $Z_1, \ldots, Z_n$.

Why bother with the extra U? This combines the running-average guarantee (Theorem 5.6) with the *uniformly randomized* Markov inequality (a sharper, randomized version of Markov's inequality introduced in Chapter 2, which can strictly increase power at the cost of injecting independent external randomness U). In the numerical studies of Section 5.7, using this randomized threshold in place of the plain Markov threshold noticeably improves power (the "UMI" / "EUMI" variants), at the modest cost of extra randomization.

Python: exchangeable Markov's inequality, demonstrated

A quick simulation confirming that the trajectory-wide guarantee (5.3) actually holds for an exchangeable sequence of e-variables (here, iid log-normal e-variables with mean 1).

```
import numpy as np

rng = np.random.default_rng(0)
alpha, n_steps, reps = 0.05, 500, 20000
sigma = 1.0
exceed_count = 0
for _ in range(reps):
    # iid e-variable: E[Z]=1 by construction (lognormal, mean-corrected)
    Z = np.exp(rng.normal(-sigma**2/2, sigma, n_steps))
    running_mean = np.cumsum(Z) / np.arange(1, n_steps + 1)
    if np.any(running_mean >= 1/alpha):
        exceed_count += 1
print(f"P(sup_n running_mean >= 1/alpha) ~= {exceed_count/reps:.4f}")
print(f"EMI bound: alpha = {alpha}")   # empirical rate should be <= alpha
```

Running this shows the empirical exceedance rate safely below $\alpha = 0.05$ – Theorem 5.6 is not just valid but often quite loose in this iid case, which is expected since (5.3) has to hold for *every* exchangeable sequence, including adversarial ones.

5.5 Intuition: from tests to confidence sets, for free

Setup. Suppose distributions are indexed by a parameter, $\{p_\theta\}_{\theta\in\Theta}$, and the true parameter is θ^*. A $(1-\alpha)$ confidence set is a data-dependent set $C(X^n)$ such that $\mathbb{P}_{\theta^*}(\theta^* \in C(X^n)) \geq 1-\alpha$ for every possible $\theta^* \in \Theta$.

The standard trick: test inversion

Test *every single point* $\theta \in \Theta$ as if it were a simple null ("$H_0 : \theta^* = \theta$") at level α, and collect all the θ's that survive (are not rejected). Because each individual test has type-I error $\leq \alpha$, the true θ^* survives its own point-null test with probability $\geq 1-\alpha$ – so the collected set covers θ^* with probability $\geq 1-\alpha$. This works for *any* valid test, with no regularity conditions – so it works for the split LRT.

Why this is attractive here: classical confidence sets usually need asymptotic normality of an estimator, or a known sampling distribution for a test statistic. Universal inference needs neither.

Constructing the universal confidence set I

Split X^n into D_0, D_1 as before. Let $\hat{\theta}$ be any estimator computed from D_1 alone. Define

$$C(X^n) := \left\{ \theta \in \Theta \,:\, \prod_{i \in D_0} \frac{p_{\hat{\theta}}(X_i)}{p_\theta(X_i)} < \frac{1}{\alpha} \right\}.$$

Notation

Write $\mathcal{L}_0(\theta) := \prod_{i \in D_0} p_\theta(X_i)$ for the (null-family) likelihood evaluated on D_0 alone. Then

$$C(X^n) = \left\{ \theta \in \Theta \,:\, \mathcal{L}_0(\hat{\theta}) / \mathcal{L}_0(\theta) < 1/\alpha \right\}.$$

In words: θ is in the confidence set exactly when the point hypothesis $\theta^* = \theta$ is *not rejected* by the split LRT that compares $\hat{\theta}$ (fit on D_1) against θ (using D_0).

Constructing the universal confidence set II

Proposition 5.8

For any $\theta^* \in \Theta$ and any predefined $\alpha \in (0, 1)$, $C(X^n)$ above is a $(1 - \alpha)$ confidence set for θ^*.

The proof is immediate from Theorem 5.1: for the true θ^*, $E = \mathcal{L}_0(\hat{\theta})/\mathcal{L}_0(\theta^*)$ is an e-variable, so $\mathbb{1}_{\{E \geq 1/\alpha\}}$ is a level-α test, and θ^* fails to be excluded from $C(X^n)$ with probability $\geq 1 - \alpha$.

Example: multivariate Gaussian mean, identity covariance

Let $X_1, \ldots, X_n \overset{\text{iid}}{\sim} N(\theta^*, I_d)$. Here $C(X^n)$ turns out to be a Euclidean ball (spherical confidence region) around $\hat{\theta}$. A natural question: what fraction p_0 of the data should go into D_0 (used for the ratio) versus D_1 (used to compute $\hat{\theta}$)?

Theorem 5.9: optimal splitting proportion

The splitting proportion p_0^* that minimizes the expected squared radius $\mathbb{E}^{\mathbb{P}}[r^2(C(X^n))]$ of the confidence set is

$$p_0^* = 1 - \frac{\sqrt{4d^2 + 8d\log(1/\alpha)} - 2d}{4\log(1/\alpha)}.$$

Reading the formula: d is the dimension, α the miscoverage level. As $d \to \infty$ (fixed α), $p_0^* \to 0.5$ (equal split is asymptotically optimal in high dimensions). As $\alpha \to 0$ (fixed d), $p_0^* \to 1$ (almost all the data should go toward estimating the null-family likelihood), but this only matters at extreme, impractical significance levels – e.g. at $d = 1$ one would need $\alpha < e^{-40}$ before p_0^* exceeds 0.90.
Comparing to the classical LRT confidence set $C^{\mathrm{LRT}}(X^n)$ (whose exact finite-sample distribution is known in this Gaussian case): with an equal split, one can show $\mathbb{E}[r^2(C(X^n))]/r^2(C^{\mathrm{LRT}}(X^n)) \to 4$ as $d \to \infty$, and $\to 2$ as $\alpha \to 0$. So the universal confidence set is a small constant factor larger than the (regularity-condition-requiring) classical one – the price of validity with zero assumptions.

5.6.1 Extension: nuisance parameters

Motivation: often we only care about part of θ^*. Write $\theta^* = (\theta_0^*, \theta_1^*)$ and suppose we want a confidence set only for $\psi^* = g(\theta^*)$, a function of interest (e.g. one coordinate, or a contrast). Let $g^{-1}(\psi) = \{\theta : g(\theta) = \psi\}$.

Projecting the confidence set

Define $B(X^n) = \{\psi : C(X^n) \cap g^{-1}(\psi) \neq \varnothing\}$. Since C already covers θ^* w.p. $\geq 1-\alpha$, and $\theta^* \in g^{-1}(\psi^*)$, it is immediate that B covers ψ^* with the same probability $\geq 1-\alpha$: this is nothing but "project the confidence set for θ^* down onto the coordinate you care about."

A cleaner formula: profile likelihood

Define the profile likelihood on D_0 as

$$\mathcal{L}_0^\dagger(\psi) := \sup_{\theta:\, g(\theta)=\psi} \mathcal{L}_0(\theta).$$

Then one can rewrite the projected set directly as

$$B(X^n) = \left\{\psi : \mathcal{L}_0^\dagger(\psi)/\mathcal{L}_0(\hat{\theta}) \geq \alpha\right\},$$

5.6.2 Extension: smoothed likelihood I

Motivation: sometimes the null-family likelihood is *unbounded* (the MLE would be infinite – e.g. a density estimation problem where a point mass at a data point gives infinite likelihood). Fix this by smoothing every density with a kernel before taking likelihoods.

Smoothed densities

Let $k(x, y)$ be a kernel with $\int k(x, y)\, dy = 1$ for every x. Define the smoothed version of any density p_θ as

$$\tilde{p}_\theta(y) := \int k(x, y)\, p_\theta(x)\, dx,$$

and the smoothed empirical density on D_0 as

$$\tilde{p}_n := \frac{1}{|D_0|} \sum_{i \in D_0} k(X_i, \cdot).$$

5.6.2 Extension: smoothed likelihood II

Smoothed MLE and smoothed likelihood

The smoothed MLE is the KL projection $\tilde{\theta}_0 := \arg\min_{\theta \in \Theta_0} \mathrm{KL}(\tilde{p}_n, \tilde{p}_\theta)$. Defining the smoothed likelihood on D_0 as $\tilde{\mathcal{L}}_0(\theta) := \prod_{i \in D_0} \exp \int k(X_i, y) \log \tilde{p}_\theta(y)\, dy$, one can check $\tilde{\theta}_0$ also maximizes $\tilde{\mathcal{L}}_0(\theta)$ over $\theta \in \Theta_0$.

The smoothed split LRT is still valid

With $\hat{\theta}_1$ any estimator based on D_1, define the **smoothed split LRT**:

$$\text{reject } H_0 \text{ if } \tilde{U}_n > 1/\alpha, \quad \text{where } \tilde{U}_n = \frac{\tilde{\mathcal{L}}_0(\hat{\theta}_1)}{\tilde{\mathcal{L}}_0(\tilde{\theta}_0)}.$$

Why this controls type-I error

The proof mirrors Theorem 5.1, with two extra ingredients: (i) $\tilde{\theta}_0$ *maximizes* the smoothed likelihood, giving the same domination inequality as before; (ii) Jensen's inequality is applied *inside* the exponential/integral to swap the order of the kernel-smoothing integral and the likelihood ratio. The final identity collapses to

$$\prod_{i \in D_0} \int \tilde{p}_\psi(y)\, dy = 1,$$

exactly as in the unsmoothed proof – smoothing changes none of the logic, only the bookkeeping.

Takeaway: smoothing is a purely technical fix for unbounded likelihoods; it does not require any new ideas beyond Theorem 5.1.

5.6.3 Extension: relaxed, powered, and conditional likelihood I

Relaxations, when the exact MLE is intractable

Suppose computing the exact null MLE (or maximizing the log-likelihood) is computationally hard – e.g. it is a combinatorial (NP-hard) optimization problem. If F_0 is any *relaxation* with $\max_\theta F_0(\theta) \geq \max_\theta \mathcal{L}_0(\theta)$ (an upper bound on the max, easier to compute – e.g. a convex relaxation of a nonconvex problem), and $\hat{\theta}_0^F := \arg\max_\theta F_0(\theta)$, then

$$T_n' := \frac{\mathcal{L}_0(\hat{\theta}_1)}{F_0(\hat{\theta}_0^F)} \;\leq\; T_n := \frac{\mathcal{L}_0(\hat{\theta}_1)}{\mathcal{L}_0(\hat{\theta}_0)},$$

so T_n' is *also* a valid e-variable (it is even smaller than the exact split ratio, hence still $\leq$ the original valid e-variable in expectation). *It suffices to upper bound the maximum likelihood under the null.* E.g. testing sparsity in a high-dimensional linear model is combinatorially hard (best subset selection), but tractable convex/quadratic-programming relaxations exist.

5.6.3 Extension: relaxed, powered, and conditional likelihood II

Power likelihood, for robustness to misspecification

Replace $\mathcal{L}$ by $\mathcal{L}^\eta$ for $0 < \eta < 1$:

$$\mathbb{E}^\theta\left[\left(\frac{\mathcal{L}_0(\hat{\theta}_1)}{\mathcal{L}_0(\theta)}\right)^\eta \middle| D_1\right] = \prod_{i\in D_0} \int p_{\hat{\theta}_1}^\eta(y_i)\, p_\theta^{1-\eta}(y_i)\, dy_i \;\leq\; 1,$$

which holds because the η-Rényi divergence is nonnegative – so power-raised likelihood ratios remain valid e-variables too.

5.7 Book's numerical studies: confidence-set geometry

Setup. 1000 iid observations from $N((0,0), I_2)$ (a single fixed sample), $\alpha = 0.1$; four confidence sets compared: classical LRT, split LRT, cross-fit LRT, and subsampling LRT ($B = 100$).

What the book's coverage-region comparison (six repeated splits of the same data) shows

- The **classical LRT** region is smallest (it exploits the exact finite-sample χ^2 distribution available only in this regular Gaussian case).
- The **cross-fit LRT** region has smaller volume than the (single) **split LRT** region (though it is not literally contained within it).
- The **subsampling LRT** region also beats the single-split region, and for large B becomes approximately spherical (the algorithmic randomness of any one split averages away).
- Across the six repeated random splits, only the split / cross-fit / subsampling regions move at all – their variation is due entirely to *algorithmic* randomness, not new data.

A companion experiment varies the split proportion p_0 at $d = 10$ and $d = 1000$: the expected squared radius is minimized near the p_0^* of Theorem 5.9, and the effect of using the optimal p_0^* (versus, say, $p_0 = 0.5$) grows more pronounced as d grows.

5.7 Book's numerical studies: power comparisons

Setup. Testing $\theta^* = 0$ vs. $\theta^* \neq 0$ (each $\theta^* = c\mathbf{1}$), $n = 1000$ observations from $N(\theta^*, I_d)$, $d \in \{1, 2, 10\}$, power plotted against $\|\theta^*\|^2$.

Ranking of the four procedures

Power increases with $\|\theta^*\|^2$, as expected, and at every fixed $\|\theta^*\|^2$ the ranking is: **classical LRT** (highest power) $>$ **subsampling LRT** $>$ **cross-fit LRT** $>$ **split LRT** (lowest power, single split). As d grows, the gap between subsampling and cross-fit widens.

Gaussian mixture model selection. Now consider

$$X_1, \ldots, X_n \overset{\text{iid}}{\sim} 0.25 \cdot N(\mu_1, 1) + 0.75 \cdot N(\mu_2, 1),$$

testing $H_0 : \mu_1 = \mu_2$ vs. $H_1 : \mu_1 \neq \mu_2$. Here the likelihood ratio statistic λ satisfies $-2\log\lambda \xrightarrow{d} \max(0, Z)^2$ for $Z \sim N(0, 1)$ – *not* the usual χ^2_1 – so the "LRT" (reject if $-2\log\lambda > q_{1-2\alpha}$ of χ^2_1) is only usable here as a numerical *benchmark*, because its exact behavior is known only when the mixing weights $(0.25/0.75)$ are assumed known; it does not generalize to unknown/general mixtures, unlike universal inference.

5.7 Book's numerical studies: mixture model selection

Six universal-inference variants are compared against the LRT benchmark ($n = 500$, 500 replications, $\mu := -\mu_1 = \mu_2 \in [0, 1]$):

- **UI**: split LR e-variable + (plain) Markov's inequality;
- **UMI-UI**: split LR e-variable + uniformly randomized Markov;
- **SUI**: subsampled LR e-variable + (plain) Markov;
- **UMI-SUI**: subsampled LR e-variable + uniformly randomized Markov;
- **EMI-SUI**: subsampled LR e-variable + exchangeable Markov (Thm 5.6);
- **EUMI-SUI**: subsampled LR e-variable + exchangeable, uniformly-randomized Markov (Thm 5.7).

Findings

UMI-UI uniformly dominates UI; likewise UMI-SUI, EMI-SUI, EUMI-SUI all dominate plain SUI (randomization / exchangeability-aware thresholds are essentially free power gains – around 15% absolute, for some μ). UMI-SUI and EUMI-SUI perform similarly. All universal-inference variants are markedly more conservative (lower power) than the LRT benchmark – the price of a test that remains valid for *arbitrary* (not just known-weight) mixture models, where the LRT benchmark simply does not apply.

Python: our own reproduction (5.7) – part I

A simplified version of the book's mixture-testing study: power of the split LRT vs. the subsampled LRT ($B = 3$) as sample size n grows, testing $H_0 : \mu_1 = \mu_2$ against a mixture with known weights $0.25/0.75$ and a fixed true gap $\mu_1 - \mu_2 = -1.6$.

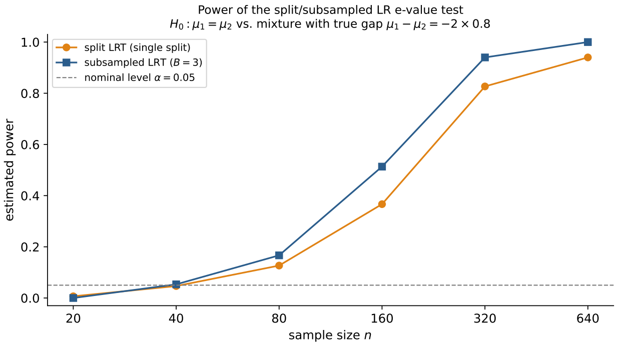
```
import numpy as np
from scipy.stats import norm
from scipy.optimize import minimize

def split_lr_logE(D0, D1, lam=0.75):
    def negloglik(p, x):
        mu1, mu2 = p
        d = (1-lam)*norm.pdf(x,mu1,1) + lam*norm.pdf(x,mu2,1)
        return -np.sum(np.log(np.clip(d, 1e-300, None)))
    res = minimize(negloglik, x0=[D1.mean()-0.5, D1.mean()+0.5], args=(D1,),
                   method="Nelder-Mead")
    mu1h, mu2h = res.x
    mu0h = D0.mean()
    q1 = (1-lam)*norm.pdf(D0,mu1h,1) + lam*norm.pdf(D0,mu2h,1)
    p0 = norm.pdf(D0, mu0h, 1)
    return np.sum(np.log(q1) - np.log(p0))
```

Python: our own reproduction (5.7) – part II

Estimating power at a given n by Monte Carlo: repeatedly simulate data, run $B = 3$ independent splits, and check both the plain split-LRT ($E^{(1)}$ alone) and the subsampled LRT (average of the B e-variables) against the threshold $1/\alpha$.

```
def power_at_n(n, mu, alpha=0.05, reps=150, B=3, lam=0.75, rng=None):
    rej_split, rej_sub = 0, 0
    for _ in range(reps):
        comp = rng.random(n) < lam
        x = np.where(comp, rng.normal(mu,1,n), rng.normal(-mu,1,n))
        logEs = []
        for _ in range(B):
            idx = rng.permutation(n); h = n//2
            logEs.append(split_lr_logE(x[idx[:h]], x[idx[h:]], lam))
        if logEs[0] >= np.log(1/alpha): rej_split += 1
        if np.mean(np.exp(logEs)) >= 1/alpha: rej_sub += 1
    return rej_split/reps, rej_sub/reps
```

Our reproduction: power vs. sample size



Results $(150$ reps per $n)$

n	split	subsampled $(B{=}3)$
20	0.007	0.000
40	0.047	0.053
80	0.127	0.167
160	0.367	0.513
320	0.827	0.940
640	0.940	1.000

Exactly as Proposition 5.3/5.4 predict: at every n, averaging $B = 3$ independent splits (subsampled LRT) matches or beats a single split, and the gap widens as n grows and power moves away from both 0 and 1 (where averaging can help the most).

Chapter 5 summary I

The reduction at the heart of the chapter

Universal inference reduces composite-vs-composite hypothesis testing, under *no regularity conditions*, to computing (or upper bounding) the maximum likelihood under the null.

- **5.1** Irregular models (unidentified mixture parameters, shape constraints like log-concavity) break classical asymptotic theory but are exactly where universal inference shines.
- **5.2** Split data into D_0, D_1; pick $\hat{q}_1$ on D_1; fit the null MLE $\hat{p}_0$ on D_0; the ratio $E = \prod_{i \in D_0} \hat{q}_1(X_i)/\hat{p}_0(X_i)$ is an exact e-variable (Theorem 5.1), because $\hat{p}_0$ dominates p exactly on the data it is tested against.
- **5.3** Averaging over many independent splits (the subsampled/cross-fit e-variable) removes algorithmic randomness and can only improve e-power (Jensen/concavity of $\log$), while remaining sequentially valid.
- **5.4** The exchangeable Markov inequality (and its uniformly-randomized version) is the technical engine that makes "keep splitting until convinced" a valid, non-cheating stopping rule.

Chapter 5 summary II

- **5.5** Inverting the split LRT point-by-point gives universal confidence sets with no regularity conditions, at the cost of a small constant-factor inflation versus classical (regularity-requiring) sets.
- **5.6** The same idea extends to nuisance parameters, unbounded likelihoods (via smoothing), and computationally hard MLEs (via relaxations and power likelihoods).
- **5.7** Across every numerical study, the ranking is consistent: classical (when available) $>$ subsampling $\approx$ cross-fit $>$ single split – but only universal inference remains valid when classical asymptotics fail.